

Lender Brief

What 20,000 loan applications tell a lender, and what to do with it

20,000 applications studied	24% of applications were approved	88% of decisions predicted correctly on cases the model had not seen
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The question

Which details about a loan applicant are linked to the loan being approved, which are not, and what should a lender do with that knowledge? The study also checks whether a model built on these details treats different groups of applicants fairly.

Three main findings

1. Affordability drives approval

When the loan is up to about a quarter of the applicant's yearly income, 69% of applications are approved. When it is more than three quarters of yearly income, fewer than 1% are. The share of income already going to debt gives the same picture: 67.5% approved among applicants with the least debt, 0.5% among those with the most.

2. Much of what lenders might weigh makes little or no difference

Savings balance and time in the current job do not change the chance of approval in any real way, because their differences are no bigger than chance would produce. Employment status and home ownership make a small difference: 18% of unemployed applicants are approved against 24% of employed and 28% of self employed ones. Credit score matters moderately, from 14% approved below a score of 450 to 45% at 650 and above.

3. A simple model is enough

A statistical model that uses only applicant information gets 88% of decisions right on applications it has never seen. A rule that rejects everyone would get 76% right. Using just two ratios, loan against income and debt against income, still gets 86% right. A more complex method (a random forest) did no better.

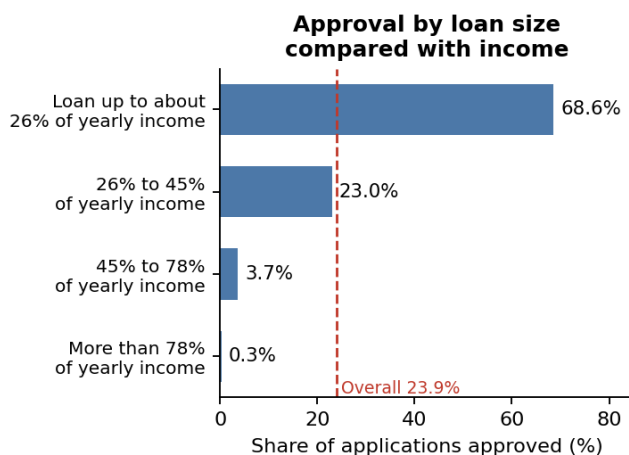


Figure 1. Share of applications approved, by size of the loan compared with yearly income. The dashed line is the overall approval rate.

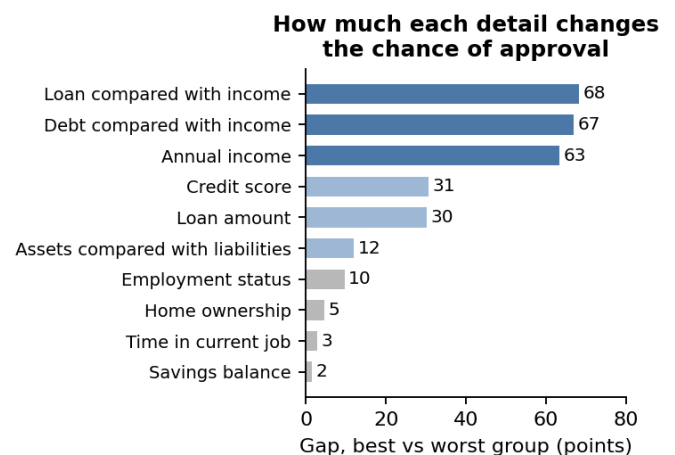


Figure 2. How many points the approval rate differs between the best and the worst group of each detail. Dark blue: very large, light blue: moderate, grey: small or none.

What a lender should do

- Judge affordability first.** Use the size of the loan against income, and debt against income, as the main checks. Both can be explained to an applicant in plain words.
- Stop weighing what does not matter.** Do not reject someone only for a short time in their job or a low savings balance. The data does not support it.
- Use the model to support people, not to replace them.** At the standard cutoff it misses about 3 in 10 of the loans that were approved. Send borderline cases (a chance of approval of about 30% to 50%) to a person.

4. **Drop details that add almost nothing.** Employment status and home ownership barely change accuracy (88.0% with or without them) and can raise fairness concerns, so consider removing them. Rules on what may be used differ by country, so get legal advice first.
5. **Check fairness regularly.** Repeat the group checks below on every new version of the model, looking both at who is approved and at how many good applications are missed in each group.
6. **Do not use the lender's own risk score or interest rate to explain decisions.** They come from the same process, so they make a model look better (98% correct) without explaining anything.

Fairness check

The model does not use age, marital status or education. Other details can stand in for them, so the table checks whether its decisions still differ between groups. It uses the 4,000 applications kept aside for testing.

Group	Applicants	Approved in the data	Model approves	Approved loans the model finds
By age group				
Under 30	773	13.3%	12.8%	60.2%
30 to 39	1,227	22.0%	18.6%	64.1%
40 to 49	1,197	28.4%	22.9%	69.4%
50 to 59	605	28.4%	25.0%	73.8%
60 and over	198	35.9%	33.8%	78.9%
By marital status				
Divorced	540	23.0%	21.7%	72.6%
Married	2,037	24.0%	21.3%	71.8%
Single	1,251	24.1%	18.7%	61.6%
Widowed	172	23.8%	20.3%	65.9%
By employment status				
Employed	3,402	24.2%	20.7%	68.8%
Self-Employed	311	26.4%	24.8%	72.0%
Unemployed	287	17.4%	13.2%	56.0%
By home ownership				
Mortgage	1,624	24.9%	22.7%	70.1%
Other	415	17.3%	14.9%	66.7%
Own	786	26.5%	22.6%	72.6%
Rent	1,175	23.1%	18.0%	63.1%

Approved loans the model finds: of the loans that were really approved, the share the model also approves. A group with a low number here has more good applications missed.

Age. Approval in the data rises from 13.5% for applicants under 30 to 30.8% for those aged 60 and over, and the model follows the same pattern even though it never sees age. Younger applicants have lower incomes and credit scores and ask for larger loans compared with income, which explains part of the gap. It does not explain all of it: among the highest earners, 49% of applicants under 30 are approved against 69% of those aged 40 to 49.

Missed good applications. The model finds fewer of the approved loans for younger (60%), single (62%), unemployed (56%) and renting (63%) applicants than for older (79%), married (72%), employed (69%) and home owning (73%) ones. The single against married gap comes from smaller groups, so treat it as a flag to check rather than a proven gap.

Education. Education level is strongly linked to approval in the data (14% for high school, 44% for doctorate) even though the model does not use it.

Limits

- The data is synthetic (computer generated). The sharp patterns describe how the data was created, not how real lenders decide, so the work must be repeated on real application data before any use.
- The findings show links, not causes.
- The fairness check covers only the details in this dataset, and some groups are small. It is not a legal assessment.
- This is an analysis project, not a lending system.